

AIDoctors: Project Report

Rayce Ramsay (1009734888) Terry Tian (1007663468)
Grant Hamblin (1009860447) Hardik Shah (1009453690)

December 2, 2025

Abstract

Drug-Drug Interactions (DDIs) represent a critical source of preventable patient harm, yet current Clinical Decision Support Systems (CDSS) rely on static, "one-size-fits-all" interaction tables that fail to account for individual physiological contexts. This lack of specificity contributes to high rates of clinician alert fatigue, potentially masking severe risks in vulnerable populations. This report introduces "AIDoctors," a machine learning-based web application designed to personalize DDI risk assessment by bridging the gap between theoretical pharmacological rules and real-world clinical evidence. The proposed system utilizes a hybrid scoring model that augments deterministic DDI table lookups with a Patient Similarity Analysis (PSA) algorithm. By querying historical Electronic Health Records (EHR) to identify adverse reaction patterns in patient cohorts with matched demographics and comorbidities, the model dynamically adjusts alert severity based on longitudinal outcomes. We detail the system architecture, which integrates a Next.js front end with a Python-based inference engine, and demonstrate how this data-driven approach addresses the limitations of mechanistic static models. The results suggest that contextualizing standard DDI warnings with practice-based evidence can significantly improve risk stratification, reduce low-value alerts, and support more precise prescribing decisions.

1 Introduction

Adverse Drug Reactions (ADRs) arising from Drug-Drug Interactions (DDIs) constitute a persistent, escalating, and preventable source of harm within modern healthcare systems. As global demographics shift towards an aging population, the prevalence of multimorbidity and the consequent polypharmacy have increased the density of pharmacological regimens [1, 19]. Recent surveillance data from the FDA Adverse Event Reporting System (FAERS) indicates that over 177,000 DDI cases were reported through early 2024, with nearly 15,000 resulting in fatalities [1]. The burden of these interactions is not merely clinical, but also economic, contributing to millions of emergency department visits and hospital admissions annually, with adverse events accounting for approximately 6 out of every 1,000 ED visits [11].

In response to this challenge, Clinical Decision Support Systems (CDSS) have become ubiquitous in electronic prescribing workflows. These systems traditionally rely on curated DDI knowledge bases—static tables derived from pharmacokinetic studies and regulatory labelling [10]. While these tables provide essential categorical severity warnings, they fundamentally lack patient-specific context. They encode interaction risk as a fixed property of a drug pair, ignoring the profound inter-individual variability in physiology, renal clearance, and genetic metabolic capacity that determines whether a theoretical interaction manifests as a clinical event [6, 12].

This disconnection between theoretical risk and clinical reality has precipitated a secondary crisis: alert fatigue. Because static systems prioritize sensitivity over specificity, they generate a high volume of low-value warnings. Consequently, clinician override rates for DDI alerts frequently exceed 90%, with providers routinely bypassing even high-severity warnings [2, 9]. Common justifications for overrides, such as "patient has already tolerated combination," suggest that clinicians prioritize empirical, longitudinal evidence over static theoretical predictions [16].

This project introduces **AIDoctors**, a machine learning-based web application designed to bridge the gap between static pharmacological evidence and dynamic clinical reality. By integrating structured DDI tables with Patient Similarity Analysis (PSA) derived from historical Electronic Health Records (EHR), the system aims to personalize risk estimation. The proposed model hypothesizes that the probability of an adverse interaction is best predicted not by looking up a binary rule, but by analyzing the outcomes

of a cohort of clinically similar patients—matched on demographics, comorbidities, and medication profiles—who have been exposed to the same drug combination [23, 25]. This approach leverages the concept of "practice-based evidence" to refine the broad warnings of "evidence-based practice."

1.1 Problem Statement

Current mechanisms for DDI alerting exhibit structural deficiencies that compromise patient safety and clinical efficiency. These limitations can be categorized into three distinct domains:

First, **the lack of physiological context in static knowledge bases.** Conventional DDI tables utilize "mechanistic static models" that fail to account for dynamic patient variables such as organ function, age-related metabolic decline, and complex multi-drug regimens [10]. A warning label appropriate for a general population often underestimates risk for frail elderly patients or overestimates it for robust younger individuals [3]. For example, an interaction that causes a moderate increase in serum concentration may be benign in a healthy adult but toxic in a patient with compromised renal reserve. This "one-size-fits-all" approach leads to a dangerous imprecision in risk communication.

Second, **the systemic failure of alert fatigue.** The low specificity of rule-based systems has resulted in widespread desensitization among prescribers. With override rates for high-priority alerts approaching 87-95% [9], the safety net intended by CDSS has effectively been dismantled. The inability of current systems to filter alerts based on clinical relevance or prior patient tolerance forces clinicians to ignore the majority of warnings, increasing the probability that truly dangerous interactions will be missed amidst the noise [13]. Research indicates that inappropriate overrides are associated with a significantly higher rate of adverse drug events, confirming that the current "override culture" is actively harmful [9].

Third, **the underutilization of longitudinal historical evidence.** Electronic Health Records contain vast repositories of "practice-based evidence"—longitudinal records of patients who have tolerated or reacted to specific drug combinations. However, traditional DDI tools are incapable of leveraging this data. They operate without memory, treating a patient who has safely taken a drug pair for a decade the same as a patient receiving it for the first time [24]. The absence of similarity-based retrieval prevents the system from learning from past outcomes to refine future predictions. Advanced methodologies such as Graph Neural Networks (GNNs) and Patient Similarity Analysis offer a pathway to unlock this latent data, yet they remain largely unimplemented in bedside tools [17, 21].

The **AIDoctors** project addresses these critical gaps by replacing the deterministic lookup model with a probabilistic, data-driven framework. By augmenting DDI tables with patient similarity metrics, the system seeks to reduce false positives, validate alerts with real-world evidence, and ultimately restore the clinical value of drug interaction warnings.

2 Related Works

The integration of Artificial Intelligence (AI) into the healthcare sector has fundamentally shifted how health professionals approach patient care and treatment planning. According to the Canadian Dental Association (CDA), AI-driven tools are now pivotal in analyzing complex patient data to support real-time decision-making[5]. These technologies allow for data-driven predictions that can alert professionals to early signs of disease (e.g., cancer screening) and optimize treatment plans by analyzing medical records against vast bodies of research.

For example, a landmark study by Hwang et al. (2019) demonstrated the diagnostic superiority of AI systems [14]. The researchers tested a deep learning-based algorithm against groups of physicians, including thoracic radiologists, in detecting major thoracic diseases on chest radiographs. The AI model consistently outperformed the human readers in classification performance, highlighting the potential for AI to enhance diagnostic accuracy in clinical settings .highlighting the potential for AI to enhance diagnostic accuracy in clinical settings (see Figure 1).

This capability not only improves the speed of care delivery but also enhances patient engagement by reducing the administrative burden on providers, allowing them to focus more on direct patient interaction.

One of the most critical applications of AI in pharmacology is the prediction of Drug-Drug Interactions (DDIs). DDIs occur when the effect of a medication is altered by the simultaneous administration of another drug, potentially leading to adverse patient outcomes, loss of therapeutic effect, or toxicity.

A review by Zhang et al. (2024) emphasizes that while manual detection of these interactions is the gold standard, it is increasingly "time-consuming and labor-intensive" [26]. The sheer volume of new pharmaceutical research makes it nearly impossible for clinicians to stay updated without computational assistance. The study confirms that utilizing advancements in AI is essential for achieving accurate forecasts of DDIs, classifying prediction tasks into undirected DDI prediction, DDI event prediction, and asymmetric DDI prediction. By automating these processes, AI can significantly improve clinical decision-making and establish optimal treatment regimens that manual curation alone cannot sustain.

Current industry standards for checking interactions rely heavily on curated databases. Tools such as the DrugBank Interaction Checker [8] serve as vital resources, providing detailed databases of known interactions. However, these tools are inherently limited by their reliance on manual updates and structured data entry. As the volume of medical literature grows exponentially, these static databases often struggle to "keep up," leading to a lag between the discovery of a new interaction in research and its appearance in clinical decision support tools. This gap leaves a window where patients may be exposed to severe, undocumented interactions.

Natural Language Processing (NLP) is a research method that has found significant success in the discovery of Drug-Drug Interactions (DDIs). Studies have shown a strong correlation between the application of NLP techniques and the identification of severe interactions. Research published in Clinical Pharmacology & Therapeutics highlights the efficacy of NLP in mining unstructured text from biomedical literature to identify interactions that have not yet been codified in standard databases [7]. By scanning thousands of research papers, case reports, and clinical notes, NLP models can surface "hidden" signals of adverse interactions, providing a layer of safety that purely database-driven tools miss.

Building upon these advancements, our project, AIDoctors, combines the reliability of existing market tools with the generative and analytical power of modern AI. We have implemented a Retrieval-Augmented Generation (RAG) pipeline that integrates over 20,000 previously recorded DDI interactions from established databases. By feeding this structured knowledge into a Large Language Model (LLM), we create a system that is not only grounded in verified data but also capable of contextual understand-

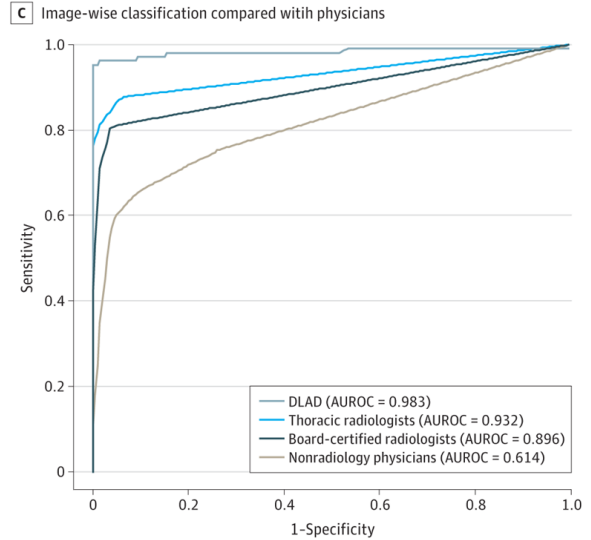


Figure 1: AI model performance in detecting thoracic diseases [14].

ing. This approach allows for a more thorough and responsive project that bridges the gap between static databases and the evolving landscape of medical knowledge.

3 Technical Approach and Design

AIDoctors integrates multiple sources of DDI data, synthetic EHR records, patient similarity analytics, and an LLM based reasoning layer into a unified system. It is organized around three components, the data engineering pipeline, the backend/inference engine, and the frontend/user interface.

Firstly, all datasets, including Synthea EHR data, AEOLUS adverse outcome signals, and multiple DDI knowledge bases (DDInter, CRESCENDDI, and Mendeley) are ingested through a unified ETL pipeline, starting with being automatically extracted via the internet. The pipeline standardizes drug identifiers using RxNorm, resolves product-level RxCUIs to ingredient level mappings (via RxNav), and aligns severity and mechanism labels across sources.

Synthea medications are transformed into structured exposure windows containing start and stop date, which enables detection of overlapping drug exposures. AEOLUS disproportionality metrics (case counts, PRR, ROR) are linked to each ingredient to generate per patient adverse event annotations. Comorbidities, demographics, and age-at-exposure are then attached to create an enriched patient-level dataset. A top k filter then selects the most clinically relevant outcome signals for each patient–drug pair.

This pipeline produces two key analytical tables: the Unified DDI Reference, a consolidated severity/mechanism mapping across the three databases, and the Patient DDI Collapsed Table, which contains all overlapping drug pairs observed in Synthea, annotated with demographics, comorbidities, AEOLUS risk signals, and whether the pair is known in external DDI sources. These tables (See Figure 2) are exported to S3 and then ingested into an RDS PostgreSQL database during deployment, forming the dataset used for patient similarity queries.

To contextualize standardized DDI warnings with real-world evidence, the backend implements a lightweight Patient Similarity Analysis. For any incoming drug pair and patient profile, the system retrieves prior cases where the same drug pair was coprescribed, the patient had similar age, the same sex, and overlapping comorbidities. These cases form a small cohort representing historically similar patients. From this cohort, the system extracts observed severities, average confidence across data sources, known mechanisms, and representative example cases. This effectively acts as a RAG (retrieval augmented generation) system, surfacing direct evidence from our corpus to enrich the model’s prediction.

The decision engine is implemented in FastAPI and integrates AWS Bedrock for generative reasoning. When a user queries an interaction, the backend retrieves the severity/mechanism from the unified DDI reference, retrieves the similar cases via RAG, and structures these into a structured prompt along with the patient specific characteristics. The LLM (gpt-oss-120b) then generates a JSON structured output separating its step-by-step reasoning from the final prediction. The API returns this prediction along with the enriched context used during inference.

The system is deployed as a cloud hosted FastAPI backend with authentication. User management and JWT-based authentication are supported through cookie-secured sessions. ETL outputs and analytics tables are persisted in AWS RDS PostgreSQL, with dataset synchronization to S3. The frontend (Next.js) interacts with the backend via the /api/predict endpoint, enabling real time queries. To see the entirety of our architecture reference figure 3.

4 Experiments and Results

This section presents the experimental evaluation of our system, detailing how we assessed its ability to generate clinically meaningful predictions and measuring the performance gains achieved through architectural, optimization, and prompt-engineering improvements.

Initially, we explored Reinforcement Learning with Verifier Rewards (RLVR) using an "LLM-as-a-Judge" framework, in which a secondary model scores the quality and correctness of each output. However, this approach proved infeasible. Our choice of verifier model (Qwen-0.6B) was too small for the task, and we lacked a sufficiently large training dataset with authoritative, ground-truth outcomes—a requirement that was unattainable within our timeline. These limitations resulted in inconsistent reward signals and poor alignment between the verifier scores and actual clinical correctness.

A key indicator of this failure was the training dynamics observed in the model’s entropy (Figure 4). We noted a rapid decrease in entropy, which typically signals that a model is becoming more confident

(less exploratory). However, qualitative analysis (as supported by other plots in Figure 4) revealed that rather than improving, the model was simply becoming "confidently wrong," converging on a suboptimal policy due to the poor quality of the training data, and low quality verifier.

Consequently, we pivoted to a deterministic evaluation method using small, curated dataset consisting of patient cases paired with known drug-drug interaction (DDI) severities and expected clinical outcomes; model predictions were compared directly against these reference labels to quantify correctness and consistency. Across this dataset, our experiments focus on three dimensions: (1) system responsiveness and latency, (2) output quality and reliability of the LLM’s clinical reasoning, and (3) the impact of retrieval quality within the RAG pipeline.

To improve system responsiveness, we concentrated on reducing the Time to First Token (TTFT), a key latency metric capturing the delay between issuing a request and the model beginning to generate its first output token. TTFT provides a sensitive indicator of bottlenecks within the RAG pipeline because it reflects the cumulative cost of preprocessing, retrieval, enrichment, and inference. As shown in Table 1, our optimizations produced a substantial reduction in TTFT, lowering the average latency from over 8 seconds to just above 5 seconds.

These gains were achieved through several targeted optimizations. First, we refined the LLM configuration by reducing the `temperature` from 0.7 to 0.15 and lowering the `max_completion_tokens` value to constrain generation length. Together, these adjustments yielded more deterministic outputs, reduced variance in completion behavior, and decreased the model’s overall compute time [15]. Second, we improved the efficiency of the data enrichment stage by parallelizing three database queries using `asyncio` with a `ThreadPoolExecutor`. Previously executed sequentially, these queries now run concurrently, achieving a 30–40% speedup. These modifications were implemented in both `app/services/prediction_service.py` and `app/routers/predictions.py`. Collectively, these system-level optimizations significantly improved end-to-end responsiveness.

Table 1: TTFT and Total Request Time Before vs After Optimization

Metric	Before (s)	After (s)	Improvement
Mean TTFT	8.154	5.091	37.6% faster
Median TTFT	8.505	5.218	38.6% faster
Min TTFT	6.632	3.129	52.8% faster
Max TTFT	9.095	6.668	26.7% faster
Std Dev	0.966	0.957	Slightly lower
Mean Total Time	8.154	5.091	37.6% faster
Median Total Time	8.505	5.218	38.6% faster
Min Total Time	6.632	3.129	52.8% faster
Max Total Time	9.095	6.668	26.7% faster
Std Dev	0.966	0.957	Slightly lower

Beyond latency, we also evaluated the accuracy and consistency of model outputs by iteratively refining our prompt design. We applied three prompt-tuning strategies that collectively improved output stability and alignment with expectations. First, we adapted the instructions to model-specific conventions and emphasized hard constraints using uppercase modal verbs (ALWAYS, NEVER), a formatting technique shown to improve instruction adherence and reduce deviation in LLM behavior [4]. Second, we enforced strict JSON output formatting by specifying an exact schema and requiring the model to return only parseable JSON, aligning with best practices in structured-output prompting [18]. Third, we rewrote task instructions as explicit, stepwise reasoning steps, which encourages the model to externalize its reasoning; chain-of-thought prompting and related structured prompting methods have been shown to improve performance on complex reasoning tasks [20]. A concise comparison of the original and improved prompts is provided in Table 2.

Table 2: Summary of Prompt Engineering Improvements

Aspect	Prompt 1	Prompt 2 (Improved)
Use of Clinical Evidence	General references to known DDIs and historical cases.	Explicit instructions to compare to known DDI severity and always reference historical cases when available.
Reasoning Structure	High-level steps for severity, comparison, cases, and concern.	Structured, stepwise reasoning with justification tied to patient-specific characteristics.
Handling Missing Data	No guidance when historical cases are absent.	Requires explicit acknowledgment of missing evidence and fallback to clinical principles.
Constraint Language	Soft instructions; no enforced constraints.	Strong constraints using uppercase (ALWAYS, NEVER) with mandatory rules.
Output Format	Suggested but not enforced; extra text allowed.	Strict JSON only: no markdown, no extra text, no trailing commas.

Finally, we assessed the impact of retrieval quality on model accuracy, as the effectiveness of a RAG system depends critically on the relevance and specificity of its retrieved context. Empirically, we found that the LLM’s output quality was highly sensitive to the similarity between the current patient and the retrieved historical cases, consistent with prior observations that overly long or non-specific context can degrade performance [22]. To address this, we implemented a custom reranking strategy that scores candidate cases based on both categorical attributes (e.g., sex, comorbidities) and quantitative differences (e.g., age). For continuous variables, we applied a decaying distance-based weighting scheme; given attribute pairs $(x_1, x'_1), \dots, (x_n, x'_n)$, the score is computed as

$$S = \sum_{i=1}^n \alpha^{i-1} |x_i - x'_i|, \quad 0 < \alpha < 1. \quad (1)$$

This reranker consistently selected more clinically relevant cases, leading to improved precision, reduced hallucinations, and stronger alignment with ground-truth drug-drug interaction patterns.

5 Future Work

- Currently our system primarily focuses on pairwise Drug-Drug Interactions (DDIs). However, clinical reality often involves patients taking three or more medications simultaneously, a phenomenon known as polypharmacy. A critical next step for AIDoctors is to implement detection capabilities for multi-drug interactions. By analyzing the combinatorial effects of multiple medications, we aim to identify complex, higher-order interactions that pairwise checks might miss, particularly for elderly populations with complex medication regimens.
- A key direction for future work is improving the quality and completeness of the data that our model depends on. Many of the challenges encountered were not due to the model itself, but to the limitations of publicly available datasets. Because Synthea and AEOLUS do not include confirmed drug to drug adverse events, hospitalization records, or detailed clinical trajectories, our system can only work with co-prescribing patterns and the severity labels provided by existing DDI databases. To address this gap we plan to implement the following changes.
 - Access to real world clinical outcomes such as lab results, symptom reports, or documented adverse events—would allow the model to learn from actual patient experiences and produce more meaningful severity estimates.
 - Improving drug normalization resources would also help, as RxNorm mappings are sometimes incomplete and required fallback heuristics during development. Using broader pharmacological vocabularies or FDA product label datasets could make drug matching more reliable.
 - Finally, richer patient records with more detailed longitudinal histories, along with additional sources beyond AEOLUS would give a fuller picture of how drug combinations behave in practice.

- Pharmacological analysis is often limited to prescription medications, yet patients frequently consume over-the-counter supplements and specific foods that alter drug efficacy. Future iterations of the project could expand the knowledge base to include Drug-Food and Herb-Drug interactions (e.g., St. John's Wort or Grapefruit juice interactions). This would require integrating distinct databases like the Dietary Supplement Label Database (DSLDB) and training the NLP models to recognize non-pharmaceutical agents in patient histories.

6 Team Reflection

The AI Doctors project was an ambitious attempt which at the beginning of this project was to modernize Medical Assistance (have an AI for a doctor) but which later on changed to tackle a more meaningful problem of Drug-Drug Interaction (DDI) checking by combining traditional interaction databases with the generative capabilities of Large Language Models (LLMs). Through the development and deployment of this platform, our team bridged the gap between theoretical AI concepts and production-grade engineering. This process provided deep insights into cloud architecture, retrieval systems, and rigorous evaluation methodologies.

A major learning curve was deploying an end-to-end cloud application using Infrastructure as Code (IaC). Moving beyond local development, we planned and developed the cloud architecture for an entire application hosted on cloud, utilizing Terraform and Docker to containerize microservices, ensuring reproducible and scalable environments. There was a significant amount of learning to be done especially everytime one of the commit crashed the deployment. Earlier access to AWS would have helped us out significantly.

We implemented Retrieval-Augmented Generation (RAG) with a database for about 20,000 previous recorded interaction, to handle personalized DDI queries relying on the database. While this pipeline proved effective, time constraints limited our opportunity to benchmark against alternative model architectures. Testing different fine-tuning or model serving techniques could have potentially yielded even better prediction outcomes or efficiency gains.

The project served as a comprehensive capstone for the course material:

1. Docker & IaC: The backbone of our deployment strategy, enabling a seamless cloud launch.
2. Prompting and Constraint Decoding: Advanced prompting and constraint decoding (e.g., forced JSON) ensured structured, machine-readable outputs critical for downstream logic.
3. Model Serving: We navigated trade-offs between external API latency and self-hosted resource needs to optimize for a responsive user experience.

In conclusion, the AI Doctors project transformed our understanding of AI engineering, transitioning us from training isolated models to orchestrating complex, cloud-native systems that solve real-world pharmacological problems.

7 Appendix

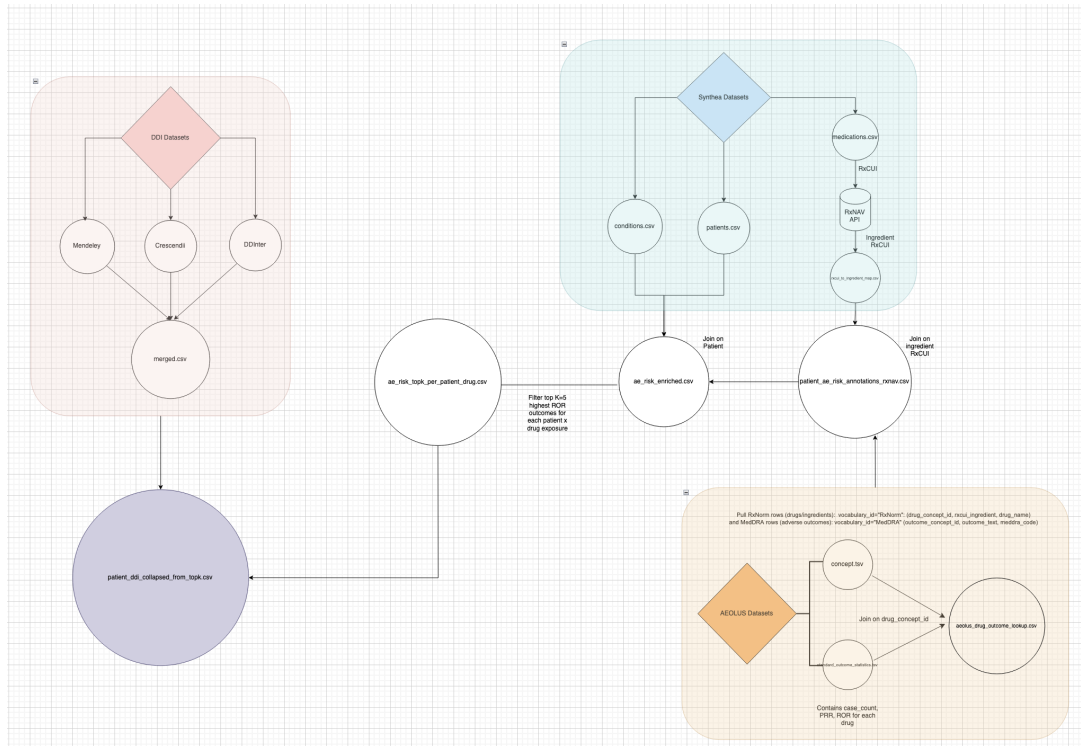


Figure 2: Data Transformation Pipeline

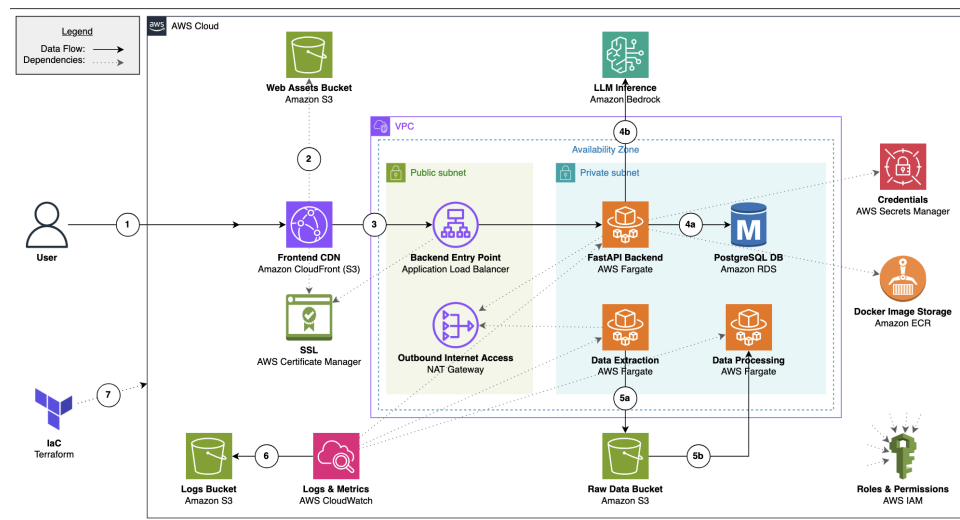


Figure 3: AWS Cloud Architecture

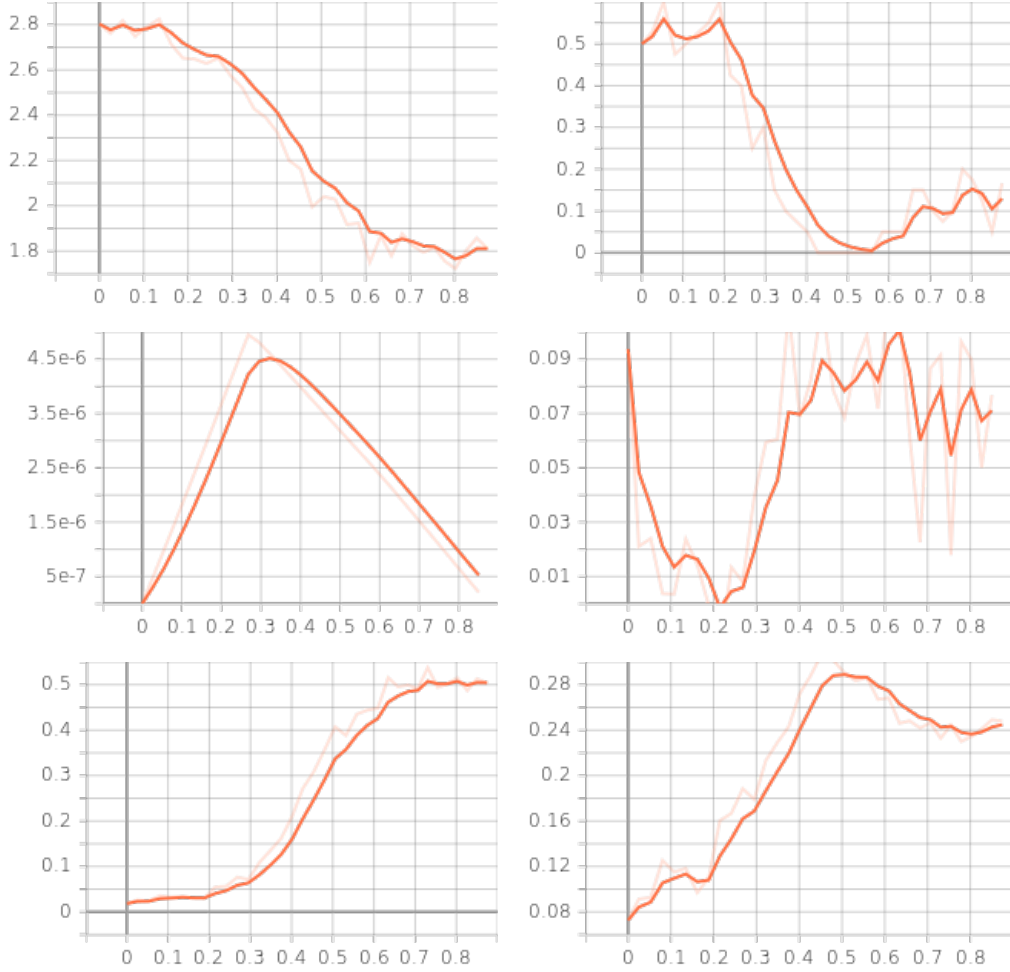


Figure 4: Training diagnostics from our failed RLVR experiment.

Top row: (Left) *Training Entropy vs. Epoch* — entropy measures uncertainty in the policy’s action distribution; rapid collapse indicates premature overconfidence. (Right) *Fraction of Zero-Variance Rewards vs. Epoch* — high values signal degenerate reward signals where the verifier assigns identical scores across samples.

Middle row: (Left) *Learning Rate Schedule* — tracks how the optimizer’s learning rate evolves throughout training. (Right) *Training Loss vs. Epoch* — expected to decrease, but instability or stagnation indicates poor optimization.

Bottom row: (Left) *Mean Severity-Specific Reward vs. Epoch* — reflects the average reward provided by the verifier model. (Right) *Reward Standard Deviation vs. Epoch* — high variance indicates inconsistent verifier scoring across episodes.

Collectively, these diagnostics show collapsing entropy, reward degeneracy, unstable loss trends, and unreliable reward signals, all of which contributed to the RLVR pipeline failing to learn a clinically meaningful policy.

References

- [1] Alahmari, A., & Fatani, S. (2025). Drug–drug interactions: A descriptive analysis of FDA adverse event reporting system. *Medicine*, 104(38), e44606. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12459552/>
- [2] Al-Jumeily, D., et al. (2020). DDI alert overrides: Rates and reasons in a tertiary hospital. *Journal of Medical Systems*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10484150/>
- [3] Al-Noor Hospital Study. (2023). Prevalence and severity of potentially serious DDIs in elderly patients. *Journal of Clinical Medicine*, 12(12), 3960. <https://www.mdpi.com/2077-0383/12/12/3960>
- [4] Arora, Akhil and Vani, Ankit and Desai, Nishkrit and Pappu, Aneesh. *Does Prompt Formatting Have Any Impact on LLM Performance?* arXiv:2406.08483, 2024. <https://arxiv.org/abs/2406.08483>.
- [5] Canadian Dental Association. (n.d.). Overview of Clinical Applications of Artificial Intelligence. <https://www.cda-amc.ca/overview-clinical-applications-artificial-intelligence>
- [6] Chazard, E., et al. (2013). Clinical significance of drug–drug interactions: The role of pathological context. *AMIA Annual Symposium Proceedings*. <https://medinform.jmir.org/2021/1/e20862/>
- [7] Discovery of Severe Previously Unknown DDI Interactions through NLP. (n.d.). *Clinical Pharmacology & Therapeutics*. <https://ascpt.onlinelibrary.wiley.com/doi/10.1002/cpt.3500>
- [8] DrugBank. (n.d.). Drug Interaction Checker. <https://go.drugbank.com/drug-interaction-checker>
- [9] Edrees, H., Amato, M. G., Wong, A., Seger, D. L., & Bates, D. W. (2020). High-priority drug–drug interaction clinical decision support overrides in a newly implemented commercial computerized provider order-entry system. *JAMIA*, 27(6), 893–900.
- [10] Einolf, H. J. (2007). Comparison of different mechanistic static models in the prediction of metabolic drug–drug interactions. *Xenobiotica*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11762507/>
- [11] U.S. Food and Drug Administration. (2024). FDA Adverse Event Reporting System (FAERS). *StatPearls Publishing*. <https://www.ncbi.nlm.nih.gov/books/NBK599521/>
- [12] Hochheiser, H., et al. (2021). Challenges in standardizing DDI data representation. *Journal of Biomedical Informatics*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12380558/>
- [13] Hussain, M. I., Reynolds, T. L., & Zheng, K. (2019). Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review. *JAMIA*, 26(10), 1141–1149.
- [14] Hwang EJ, Park S, Jin K, et al. (2019). Development and Validation of a Deep Learning–Based Automated Detection Algorithm for Major Thoracic Diseases on Chest Radiographs. *JAMA Netw Open*, 2(3):e191095.
- [15] Nguyen, V., Gao, W., Apsey, E., Kudleppanavar, G., Shah, N., and Bermudez, E. *LLM Inference Benchmarking: Fundamental Concepts*. NVIDIA Technical Blog, 2025. Available at: <https://developer.nvidia.com/blog/llm-benchmarking-fundamental-concepts/>.
- [16] Payne, T. H., et al. (2014). Override rates of drug–drug interaction alerts in electronic health records. *PLOS ONE*, 9(1). <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0085071>
- [17] PreciseADR Framework. (2023). Precision Adverse Drug Reactions Prediction with Heterogeneous Graph Neural Network. *ResearchGate*. <https://www.researchgate.net/publication/386422896>
- [18] Touvron, Hugo and Martin, Louis and others. *Structured Prompting: A Practical Guide for JSON and Programmatic Outputs in LLMs*. Industry whitepapers and structured-output best practices, 2023–2024. (No canonical arXiv version; widely referenced in JSON prompting documentation.)
- [19] Wallerstedt, S. M., et al. (2018). Prevalence and incidence of polypharmacy in the elderly: A nationwide register-based study. *Clinical Epidemiology*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5856059/>

- [20] Wei, Jason and Wang, Xuezhi and Schuurmans, Dale and Bosma, Maarten and Chi, Ed and Le, Quoc and Zhou, Denny. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. arXiv:2201.11903, 2022. <https://arxiv.org/abs/2201.11903>.
- [21] Wu, J., et al. (2023). Dual attention and patient similarity network for drug recommendation. *Bioinformatics*, 39(1), btad003.
- [22] Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A. Huerta, and Hao Peng. *Context Length Alone Hurts LLM Performance Despite Perfect Retrieval*. arXiv:2510.05381, 2025. <https://arxiv.org/abs/2510.05381>.
- [23] Zhang, A. (2018). Patient Similarity Learning and its Applications in Healthcare. *CEWIT Conference Proceedings*. <https://www.stonybrook.edu/commcms/cewit/conference2018/speakers/Aidong.pdf>
- [24] Zhang, P., et al. (2014). Tailoring treatments to individual patients based on their likelihood to respond to a therapy. *AMIA Summits on Translational Science Proceedings*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4333693/>
- [25] Zhang, Y., et al. (2023). Deep Dynamic Patient Similarity Analysis Model Development and Validation in ICU. *Frontiers in Medicine*. <https://www.researchgate.net/publication/362149837>
- [26] Zhang, Y., Deng, Z., Xu, X., Feng, Y., & Junliang, S. (2024). Application of Artificial Intelligence in Drug-Drug Interactions Prediction: A Review. *Journal of Chemical Information and Modeling*. <https://pubs.acs.org/doi/10.1021/acs.jcim.3c00582>